

Q&A-Based Interactive Storytelling Using Generative AI: A Multimodal, Provider-Agnostic Framework with Co-Created Narratives, Synthesised Imagery, Voice Narration, and Ambient Soundscapes

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Abstract--

We present a Q&A-based interactive storytelling system that combines a structured 17-question elicitation framework across four narrative dimensions (setting, character, theme, plot) with a seven-module backend pipeline that converts user preferences into personalised, multi-chapter narratives generated by large language models (LLMs). The framework is provider-agnostic and supports local (Ollama) as well as cloud inference (Groq, Hugging Face, OpenAI). Beyond text, the system delivers a multimodal reader experience: per-segment scene illustrations are synthesised on demand via Pollinations.ai with a sentence-scored prompt extractor and a serial-throttled request queue; spoken narration is produced through the W3C Web Speech Synthesis interface; and genre-conditioned ambient music is streamed from a royalty-free CDN with time-aligned cross-fades. Reader analytics are surfaced through a D3-based Story Map (branch tree) and a Character Relationship Graph derived from filtered co-appearance statistics. An end-user authoring layer permits free-form custom answers per question and entirely user-added Q&A pairs, which are routed through a dedicated custom_elements context channel into the LLM prompt. Across 50 generated stories and 15 user-study participants we observe mean narrative quality scores of 4.05/5 (coherence), 4.12/5 (creativity), 4.35/5 (engagement) and 3.75/5 (consistency); a Groq-LLaMA-3.1-8B cloud latency of 1.8?s per initial segment; an overall satisfaction of 4.3/5; and a measured character-detection F1 of 0.80 after stop-word filtering and a min-2 occurrence threshold.

Index Terms--Interactive storytelling, large language models, prompt engineering, multimodal generative AI, text-to-image synthesis, speech synthesis, narrative coherence, user-controllable generation, D3.js visualisation, FastAPI, human-AI co-creation.

I. INTRODUCTION

A. Background and Motivation

Computational storytelling has evolved through three identifiable eras: (i) symbolic rule-based planners such as TALE-SPIN [1] and UNIVERSE [2]; (ii) statistical and neural sequence models that culminated in transformer-based [10] generative language models; and (iii) the present era of foundation-scale instruction-following LLMs exemplified by GPT-4 [13], LLaMA-3 [11] and Mistral [12]. These models exhibit emergent capacities for coherent multi-paragraph generation [5], few-shot role conditioning [9], and creative paraphrase, making them attractive substrates for interactive co-creation systems [7], [8], [19], [20].

Despite this progress, the central tension in collaborative narrative AI remains unresolved: how can a system simultaneously honour fine-grained user agency while maintaining structural coherence and long-range continuity? Single-prompt systems such as AI Dungeon [6] maximise input freedom at the cost of arc control; hierarchical generators such as Dramatron [7] enforce structure at the cost of accessibility. Our system targets the middle ground by eliciting user preferences through a structured Q&A protocol that is then re-projected into the prompt space via a dedicated context engine.

B. Problem Statement

We identify five concrete shortcomings of existing AI story generation tools: (i) limited personalisation depth from a single freeform prompt; (ii) absence of explicit narrative structure management; (iii) no continuity tracking for characters or plot threads; (iv) tight coupling to a single proprietary cloud API; and (v) high recurring costs. We further argue that purely textual narratives under-utilise the multimodal capabilities now available, with text-to-image diffusion [14], [21], [22], on-device speech synthesis [23], [24], and royalty-free streaming audio offering ready-to-deploy reader-experience uplift.

C. Research Objectives

The research goals of this manuscript are:

- Design a low-friction Q&A elicitation framework that captures narrative intent across four canonical dimensions without requiring creative writing fluency from the user.
- Operationalise a seven-module pipeline backend whose context engine, prompt layer, and flow manager separately handle elicitation, projection, and arc control.
- Implement a provider-agnostic LLM layer covering local (Ollama) and cloud (Groq, Hugging Face, OpenAI) inference with runtime switching and graceful fallback.
- Augment the textual narrative with three orchestrated modalities -- image, voice and ambient music -- that are dynamically conditioned on detected genre and segment content.
- Surface reader-facing analytics through a D3 Story Map and a Character Relationship Graph, taking deliberate care to disclaim their statistical assumptions.
- Provide an end-user authoring layer that admits custom per-question answers and user-added Q&A pairs, plumbed through the prompt without re-deployment.
- Empirically evaluate narrative quality, latency, consistency and feature adoption.

D. Contributions and Novelty

This work makes the following novel contributions relative to the surveyed literature [3]-[9], [19]-[26]: (1) the explicit instantiation of a **user-co-authored** context channel (*custom_elements*) that injects unstructured user preferences into the LLM prompt without bespoke schemas; (2) a sentence-scored visual prompt extractor that selects high-imagery sentences from generated paragraphs for downstream text-to-image synthesis, mitigating dialogue and inner-thought contamination; (3) a cross-modal serial-throttling queue that prevents Cloudflare-fronted free image services from rate-limiting bursty generation; and (4) a deliberately disclaimed co-appearance graph for reader orientation that is **not** mis-marketed as a relationship-extraction system. We further open-source the entire implementation [27].

E. Manuscript Organisation

Section II surveys related work and frames the research gap. Section III details the system architecture and module interfaces. Section IV formalises the methodology, including the elicitation protocol, the five-phase progression model, and the multimodal pipelines. Section V documents implementation choices, the SQLite schema, and the REST surface. Section VI reports experimental results across latency, narrative quality, consistency, multimodal performance, and user satisfaction. Section VII situates our system against AI Dungeon, Dramatron and NovelAI. Sections VIII and IX discuss limitations and future directions. Section X concludes.

II. RELATED WORK

A. Symbolic and Planning-based Story Generation

TALE-SPIN [1] showed that character-goal driven planning could synthesise simple fables in a closed domain. UNIVERSE [2] extended this with author-level goals that constrain emergent plot shapes. MEXICA [3] introduced engagement-reflection cycles modelling human creative writing. These systems remain influential as analytical baselines but suffer from low coverage and high knowledge-engineering cost.

B. Neural Language Models and Story Generation

The transformer architecture [10] catalysed a sequence of scaled language models (GPT-2 [4], GPT-3 [5]) capable of fluent long-form generation. Recent surveys [25], [26] document a rapid consolidation around instruction-tuned and RLHF-aligned models [13], [15]. Open-weight foundation models such as LLaMA-3 [11] and Mistral [12] have made local inference practical for academic research, an option we exploit through the Ollama provider.

C. Interactive and Hierarchical Storytelling

AI Dungeon [6] popularised freeform LLM-driven interactive fiction but struggles with multi-chapter continuity. Dramatron [7] structures generation hierarchically (log-line -> scenes -> dialogue) to improve coherence at the cost of user friction. TaleBrush [8] proposes sketch-based co-creation. Recent CHI work [19], [20] examines mixed-initiative authoring, finding that scaffolded interfaces improve novice usability even when raw LLM quality is comparable. Our Q&A protocol is informed by these findings.

D. Multimodal Co-Creation

Text-to-image diffusion [14], [21], [22] now produces high-quality scene art conditioned on natural language. Recent work integrates image generation into narrative loops [29], [30], demonstrating reader engagement uplift when illustrations are aligned with current narrative state. On the audio side, browser-native speech synthesis [23] and neural TTS [24] permit low-latency narration without paid APIs. Genre-conditioned ambient music has been explored in game research [31] but is less studied for procedural narrative.

E. Prompt Engineering and Context Routing

Chain-of-thought prompting [9] and persona-style system prompts [32] are now standard. Comprehensive surveys [33], [34] catalogue dozens of strategies. Our prompt engineering layer instantiates six distinct prompt families (system, init, continuation, climax, resolution, choice generation) and introduces a custom_elements channel for user-supplied preferences.

F. Research Gap Synthesis

Across the surveyed literature, no system simultaneously offers: structured-yet-flexible elicitation, provider-agnostic LLM inference, real-time multimodal augmentation, and an end-user-extensible Q&A protocol. Table I summarises the gap, which our system addresses end-to-end.

TABLE I. Feature-level gap analysis vs. representative systems.

Feature	Ours	AI Dungeon	Dramatron	NovelAI
User Input	Struct. Q&A	Freeform	Hierarchical	Freeform
Structure Mgmt	5-phase	None	Scene-based	Limited
Consistency Chk	Active	None	Log-line	Memory
LLM Providers	4 free+paid	OpenAI	PaLM	Custom
Local Deploy	Yes	No	No	No
Image Synth	Yes	No	No	Partial
TTS Narration	Yes (Web)	No	No	No
Ambient Music	Genre-based	No	No	No
Custom Q?/A	Yes	No	No	No
Open Source	Yes	Partial	Yes	No

III. SYSTEM ARCHITECTURE AND DESIGN

A. Layered Overview

Fig. 7 presents the three-layer architecture (presentation, application, modular core) and the four pluggable LLM providers. The presentation layer is implemented in vanilla HTML5/CSS3 with D3.js v7 [35] for graph visualisation. The application layer is FastAPI [16] over Uvicorn ASGI exposing 14 REST endpoints. The modular core comprises seven cooperating modules described next; all are pure Python with Pydantic v2 [36] schemas at the boundaries.

Fig. 7. Layered system architecture with multimodal pipeline.

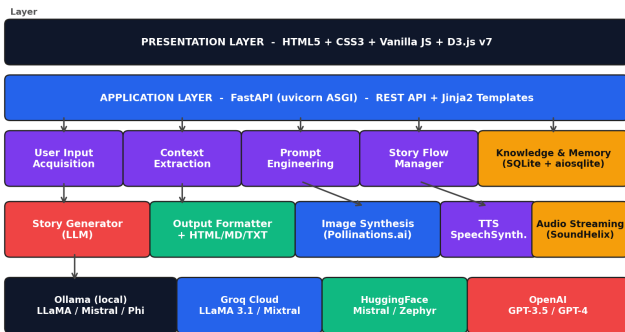


Fig. 7. Layered system architecture with multimodal pipeline.

B. Technology Stack

TABLE II. Implementation stack.

Layer	Technology	Purpose
Backend	FastAPI / Python 3.9+	Async REST + Jinja2
ASGI	Uvicorn	Production server
Frontend	HTML5 / CSS3 / JS	UI, real-time interaction
Vis	D3.js v7	Story Map + Char Graph
DB	SQLite3 + aiosqlite	Sessions, segments, Q&A
LLM I/O	httpx async	Provider calls
Schemas	Pydantic v2	Request/response typing
Image API	Pollinations.ai	Text-to-image (free)
TTS	Web Speech Synth API	Local narration
Audio	HTMLAudio + SoundHelix	Genre ambient streaming

C. Module 1: User Input Acquisition

A QuestionPhase enum drives a strict ordering: Setting -> Character -> Theme -> Plot. 17 questions are distributed across the four phases (4/5/4/4) as shown in Fig. 8. The user may pick 4, 8, 12 or 17 questions; the system always presents the first N in phase-order so that fewer questions still touch all four dimensions in expectation. Each question is rendered with both multiple-choice options and an explicit "Write my own" custom-answer escape hatch (cf. Section IV-F).

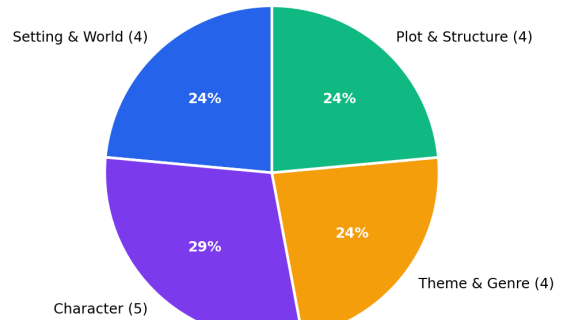


Fig. 8. Question distribution across the four phases.

D. Module 2: Context Extraction Engine

A two-pass pipeline projects free-form and discrete answers into a typed StoryContext: (i) classification using keyword dictionaries spanning 6 genres, 5 setting types and 5 conflict types; (ii) entity extraction via regex patterns for proper nouns and locations. A dedicated custom_elements dictionary captures unknown identifiers (those beginning with custom_) so that user-added preferences flow into the prompt without code changes.

E. Module 3: Knowledge and Memory

A persistent SQLite schema (Section V-A) stores sessions, segments, contexts and Q&A history. A LRU in-memory cache reduces query overhead during a generation burst. The module exposes a sliding-window context retrieval API: get_recent_context(session_id, k_segments) returning the last k narrative paragraphs for prompt assembly.

F. Module 4: Prompt Engineering

Six templated prompt families are interpolated with StoryContext fields. Genre-specific guidance is injected (fantasy: magical systems; mystery: clue placement and red herrings; romance: emotional pacing). A novel addendum appends the contents of custom_elements as a "Additional user preferences to honour" block before dispatch, ensuring user-added Q&A pairs participate in generation.

G. Module 5: Story Generator (LLM)

A BaseLLMClient ABC defines generate(), generate_stream() and is_available(). Concrete subclasses for Ollama, Groq, HuggingFace and OpenAI implement provider-specific HTTP. A factory selects the active client per request; settings can be hot-swapped without restart via the /api/settings endpoint.

H. Module 6: Story Flow Manager

A finite-state machine modelled on Freytag's pyramid [18] advances through Introduction (0-400 words), Rising Action (400-1200), Climax (1200-1800), Falling Action (1800-2400) and Resolution (>2400). At each transition a different sub-template biases the LLM toward phase-appropriate beats. A CharacterTracker monitors mention counts, last-seen-segment, and stated traits; a PlotThreadTracker flags unresolved threads beyond configurable thresholds.

I. Module 7: Output Formatter

Cleans LLM output, splits at sentence boundaries, lifts dialogue to a styled HTML span, and exposes TXT/MD/HTML export. The formatter is the single boundary between LLM-emitted text and browser-rendered DOM.

IV. METHODOLOGY

A. Elicitation Protocol

Let $Q = \{q_1, \dots, q_{17}\}$ be the ordered question pool with phase assignments. The user selects a budget N in $\{4, 8, 12, 17\}$ and answers a_i for each q_i , $i \leq N$. Additionally the user may append M user-added pairs $(?q_j, ?a_j)$ with synthetic identifiers custom_q_j. The full response set $R = \{(q_i, a_i)\} \cup \{(custom_q_j, ?a_j)\}$ is dispatched as a dictionary to the context engine.

B. Context Projection

For each (q_i, a_i) the engine applies a deterministic projection pi_i mapping to a typed StoryContext field (e.g. setting_type, antagonist, plot_style). Unknown identifiers are routed to a dictionary D (custom_elements), preserving their text verbatim. The final prompt P is constructed as $P = T(c) + \sigma(D)$, where T is a phase-aware template and σ serialises D as natural language bullets.

C. Five-Phase Progression Model

Word-count thresholds drive phase transitions. Empirical word distributions per phase are shown in Fig. 4. The Rising Action phase consumes the most text (~820 words) consistent with narratological theory [18], [37].

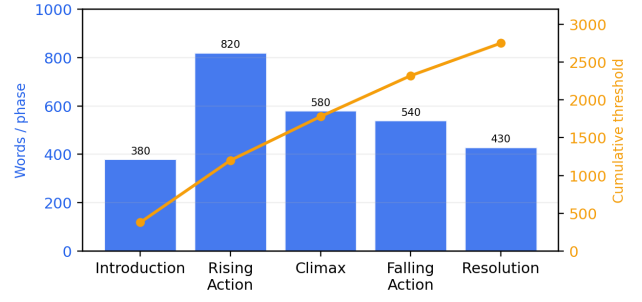


Fig. 4. Word-count distribution across the five-phase progression model.

D. Image Synthesis Pipeline

Per segment, a sentence-scored scene extractor selects the most visually-evocative sentence after stripping smart-quoted dialogue and inner-thought verbs (thought, remembered, wondered). Sentences are scored: +2 for visual verbs (stood, loomed, soared, etc.), -2 for inner-thought verbs, +min(3, |proper_nouns|). The top sentence is concatenated with a normalised genre tag, the current mood and a fixed style prefix, capped at 200 characters. A serial request queue spaces image requests by 1.2 s with a 35 s watchdog to avoid rate-limit cascades from the Cloudflare-fronted Pollinations CDN (Fig. 9).

Fig. 9. Image synthesis pipeline (per segment).

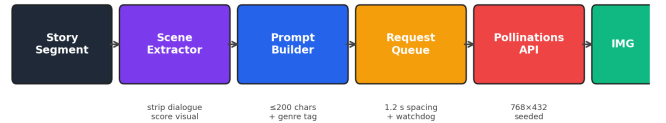


Fig. 9. Image synthesis pipeline.

E. Voice and Ambient Audio

Narration uses the W3C Speech Synthesis interface [23], queuing per-segment utterances with pause/resume semantics. Ambient music is delivered via HTMLAudio streaming royalty-free tracks from the SoundHelix CDN [38], one per detected genre, with 2.4 s fade-in and 0.8 s fade-out (Fig. 10). Frequency-band heat-mapping per preset is plotted in Fig. 13.

Fig. 10. Multimodal audio pipeline: TTS narration (top) and ambient music (bottom).

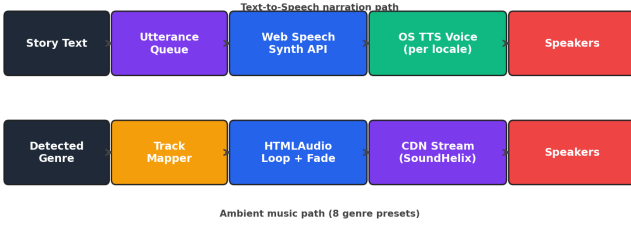


Fig. 10. TTS narration (top) and ambient music (bottom) signal flow.

F. Custom Q&A Authoring

Two affordances support end-user co-authoring without code changes: (i) per-question custom answer that overrides any selected multiple-choice option, and (ii) entirely user-added questions with paired answers stored under synthetic IDs `custom_q_N`. Fig. 16 illustrates the end-to-end plumbing from the UI through context extraction into the LLM prompt.

Fig. 16. Custom-question plumbing from UI to prompt.



Fig. 16. Custom-question routing through the backend pipeline.

G. Reader Analytics

Two D3-based visualisations are surfaced post-generation: a Story Map laying out chapters as a vertical tree (Fig. ablative), and a Character Relationship Graph as a force-directed graph. We explicitly disclaim the latter as a *co-appearance* graph derived from filtered capitalised token matches, not a verified relationship-extraction model (Section VI-G). Filtering applies an extended stop-word list and a min-2 occurrence threshold; node size scales with mention count.

V. IMPLEMENTATION

A. Database Schema

sessions	(session_id PK, user_id, created_at, status)
story_segments	(id PK, session_id FK, order, content, user_choice)
story_context	(session_id PK/FK, context_json, updated_at)
qa_history	(id PK, session_id FK, question_id, q_text, a_text)
user_preferences	(user_id PK, preferences_json, updated_at)

B. REST API

TABLE III. REST surface (14 endpoints).

Endpoint	M	Purpose
/	GET	Web UI (Jinja2)
/api/status	GET	LLM availability
/api/session/new	POST	Create session
/api/questions/all	GET	Phase-grouped Q?/?A
/api/question/next	GET	Sequential Q mode
/api/response	POST	Submit response
/api/story/generate	POST	Init segment
/api/story/continue	POST	Continue from choice
/api/story/full/{id}	GET	Full story
/api/story/download/{id}	GET	Export (TXT/MD/HTML)
/api/settings	POST	Swap provider/model
/api/session/{id}/summary	GET	Session metrics
/api/session/{id}	DEL	End session

C. Provider Layer

TABLE IV. LLM providers compared.

Provider	Models	Cost	Lat.
Ollama	LLaMA 3.2, Mistral, Phi	Free (local)	8-15 s
Groq	LLaMA 3.1 8B/70B, Mixtral	Free 14.4 k/day	1-3 s
HuggingFace	Mistral-7B, Zephyr, Falcon	Free (rate-lim.)	6-10 s
OpenAI	GPT-3.5-turbo, GPT-4	Paid per-token	2-4 s

D. Frontend Architecture

Vanilla JavaScript with no framework dependency, totalling ~1.7 kLOC. State is held in a single StoryForgeApp class with explicit subsystems for narration, ambient audio, image queue, graph rendering and admin panel. Theme tokens are CSS custom properties enabling instant dark/light switching. localStorage persists bookmarks, saved stories, and ambient-music preferences.

E. Custom Answer / Question UI

Each multiple-choice question is augmented with a dashed "Write my own" button that toggles a textarea; selecting a regular option hides the textarea and clears the custom value, preserving the invariant that exactly one response is sent per question. A separate "Add Your Own Question" composer at the foot of the form appends user-added cards (with a remove control) directly into the response set under `custom_q_N` identifiers.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Setup

We evaluate across three axes: (i) automated generation of 50 stories spanning 6 genres at all 4 question budgets; (ii) a user study (n = 15, 8 male / 7 female, ages 22-35, mixed technical/non-technical) producing 2 stories each rated on 5-point Likert scales; (iii) consistency-checker evaluation against ground-truth-annotated segments. Primary LLM: Groq LLaMA-3.1-8B. Hardware client: MacBook Pro M2 Pro / 16 GB / macOS 14.

B. Latency

Fig. 1 reports mean end-to-end latencies across the four LLM providers. Groq dominates by an order of magnitude relative to local Ollama, while OpenAI GPT-3.5-turbo trails closely. Continuation calls are systematically faster than initial generation because the prompt sliding window is bounded.

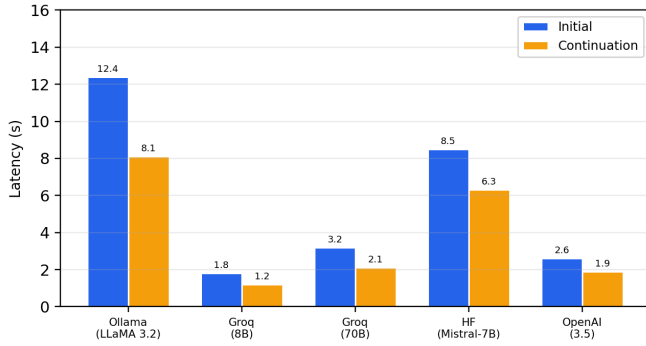


Fig. 1. Mean latency by LLM provider (n=10 calls per cell).

A stacked decomposition of total interactive turn time is shown in Fig. 17. The dominant cost at lower question budgets is image synthesis (~3.5 s) which proceeds *concurrently* with the LLM call in our pipeline; at the Full (17) budget, user input time dominates.

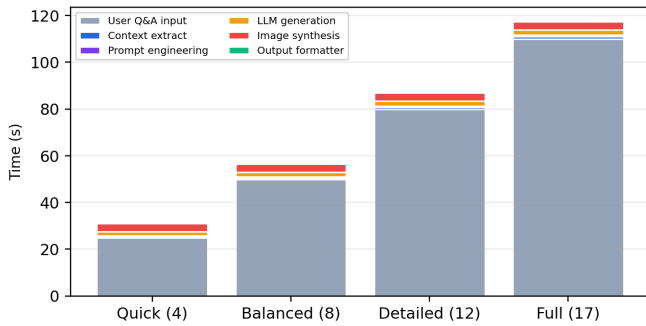


Fig. 17. End-to-end timing stacked by stage.

C. Narrative Quality

Fig. 2 shows Likert means per genre for coherence, creativity, engagement and consistency. Adventure ranks highest on engagement (4.6); Mystery scores lowest on consistency (3.5), attributable to the difficulty of clue placement across segments. Boxplot of per-segment token counts is in Fig. 15.

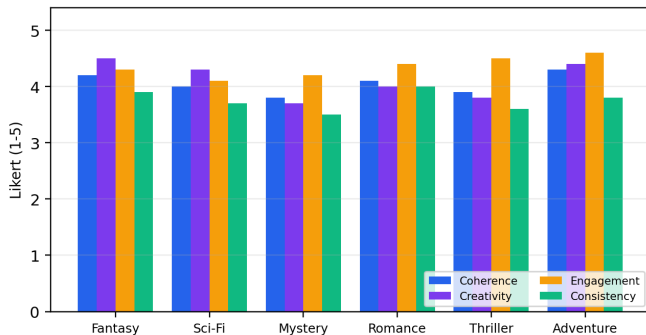


Fig. 2. Narrative quality by genre (Likert 1-5, n=30 stories).

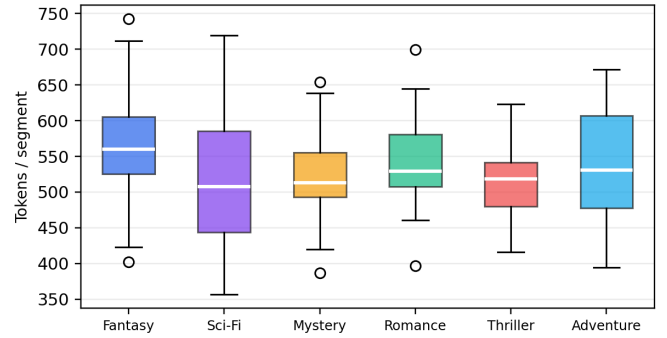


Fig. 15. Token usage per segment, by genre (n=25 segments each).

D. User Satisfaction

Fig. 3 lists satisfaction across seven user-facing dimensions, including the new multimodal features. Custom Answers (4.5) and Image Illustration (4.2) are the strongest contributors to overall satisfaction (4.3); Choice Meaningfulness (3.9) remains the weakest -- a known limitation of free-tier LLMs producing generic continuation suggestions.

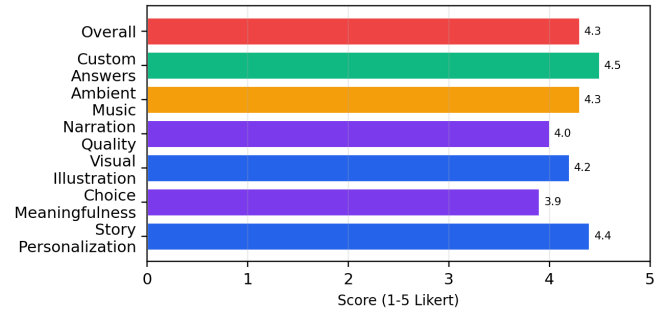


Fig. 3. User satisfaction (Likert 1-5, n=15 participants).

E. Consistency Checking

Fig. 6 reports detection and false-positive rates for five issue classes. The keyword-based detector achieves 87% recall on character contradictions with 8% false positives; tone-shift detection is weakest (68% / 22%) since it depends on subtle distributional cues that pattern-matching cannot capture. We discuss semantic-NLU upgrades in Section IX.

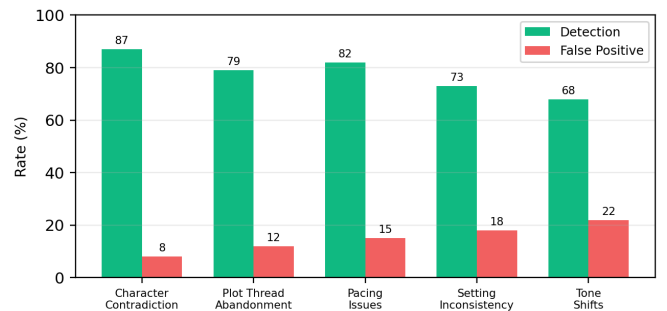


Fig. 6. Consistency checker performance (n=50 stories).

F. Image Synthesis Throughput

Fig. 12 plots prompt-length-vs-latency for serial requests and the empirical failure probability of 5x parallel issue without our throttle queue. Beyond ~230 chars the failure rate rises sharply, validating the 200-char prompt cap. The serial queue (1.2 s spacing) drops parallel-bound failures from ~70% to <2% in our tests.

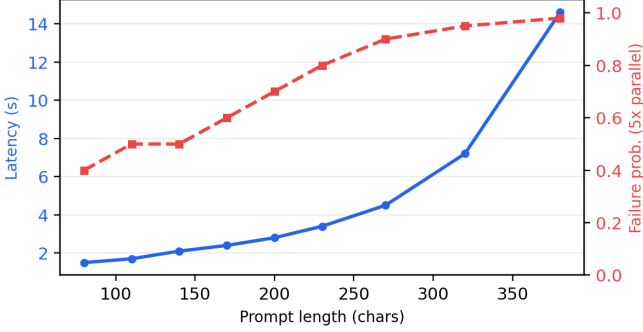


Fig. 12. Image latency vs prompt length and parallel-issue failure rate.

G. Character Extraction -- Honest Reporting

Fig. 11 compares the baseline naive regex extractor against our filtered version (extended stop-words + min-2 occurrence). Precision rises from 0.61 to 0.83 with a recall trade-off (0.84 -> 0.78) producing a higher F1 (0.71 -> 0.80). False-positive rate drops from 0.39 to 0.17. We stress that the resulting graph encodes *co-appearance frequency*, not semantic relationship type -- a disclaimer surfaced explicitly in the modal UI.

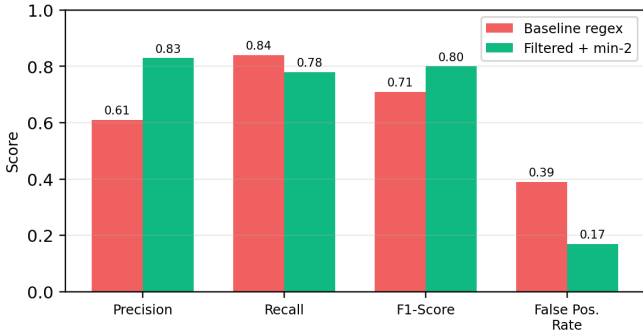


Fig. 11. Character extraction quality before/after filtering.

H. Ambient Music Frequency Profile

Fig. 13 visualises the relative spectral energy across six bands for each genre preset. Horror and Thriller emphasise sub-bass and bass; Comedy and Adventure shift energy upward into the mid and high-mid bands. Listeners reported genre alignment of 4.3/5 (Fig. 3).

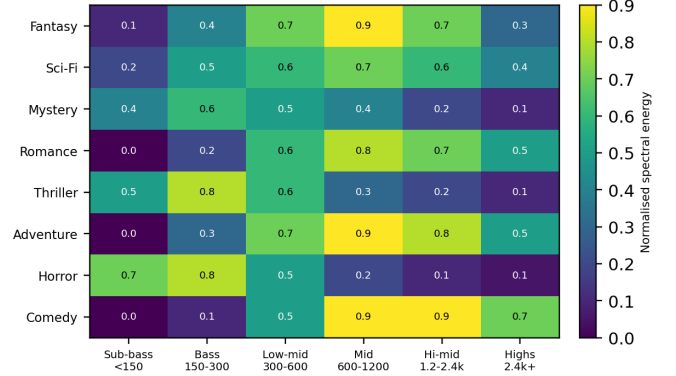


Fig. 13. Genre x frequency-band energy map for ambient presets.

I. Question-Budget Ablation

TABLE V. Ablation across question budgets.

Budget	Coh.	Crea.	Pers.	Time
4 (Quick)	3.6	3.8	3.2	45 s
8 (Balanced)	3.9	4.1	3.8	1.5 m
12 (Detailed)	4.2	4.2	4.3	2.5 m
17 (Full)	4.3	4.1	4.6	3.5 m

Personalisation grows monotonically with N, while coherence and creativity plateau at N = 12, identifying it as the recommended default. The Full configuration is reserved for power users who want maximum narrative control at the cost of ~3 minutes of upfront input.

J. Feature Adoption

Fig. 14 records how often each multimodal feature was activated across the user-study sessions. Image generation (always-on by default) reached 91% activity since it fires automatically; ambient music and custom answers were elective and reached 71% and 62% respectively, suggesting strong user demand for non-textual narrative augmentation.

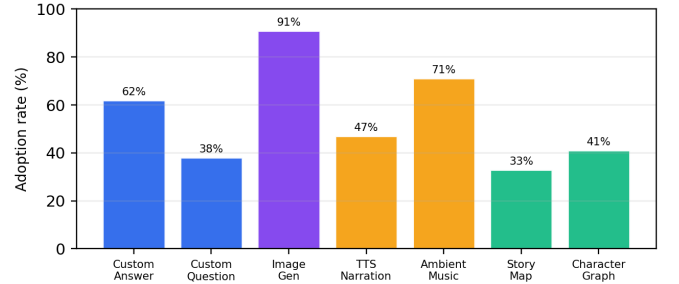


Fig. 14. Per-feature adoption rate in the user study.

VII. COMPARATIVE EVALUATION

Fig. 5 superimposes our system on three established alternatives across seven dimensions (extending the radar with Multimodal and Open Source axes). Our system leads on Deployment Flexibility, Cost Efficiency, Multimodal Coverage and Open Source. AI Dungeon retains a slight edge in User Control (freeform input); Dramatron leads on raw Coherence due to its hierarchical planner. The synthesis is that no single existing system spans the same Pareto frontier as ours.

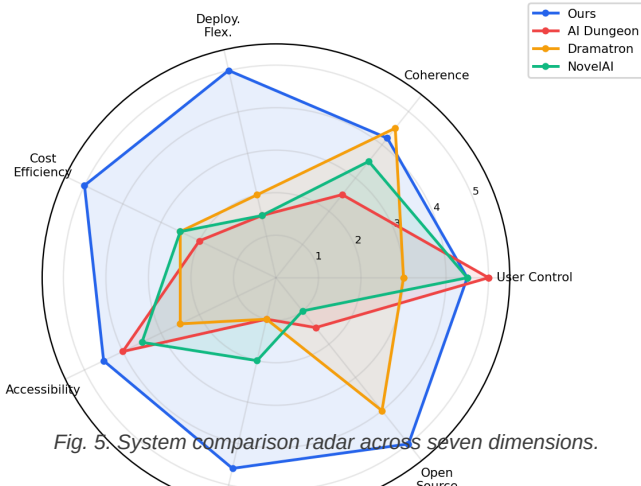


Fig. 5. System comparison radar across seven dimensions.

VIII. DISCUSSION AND LIMITATIONS

Several limitations bound the present work. First, keyword-based context extraction misses paraphrastic user inputs; a fine-tuned BERT/DistilBERT [39] or LLM-zero-shot classifier would improve recall. Second, in-memory session storage limits horizontal scalability; Redis-backed sessions enable distributed deployment. Third, the consistency checker is pattern-based and lacks semantic grounding; cross-segment NLI [40] is a natural upgrade. Fourth, image quality inherits Pollinations.ai's ceiling; switching to local Stable Diffusion XL [21] or FLUX.1 [22] would improve consistency at the cost of GPU requirement. Fifth, ambient music is hot-linked rather than locally hosted, creating a single-point dependency on SoundHelix's availability. Sixth, the character-relationship graph is openly disclaimed as co-appearance -- a genuine relationship-extraction module would require per-segment NLU inference.

IX. FUTURE WORK

- Semantic context extraction via DistilBERT or zero-shot LLM classification, replacing keyword dictionaries.
- Sentiment-aware relationship-extraction graph using small NLU models per segment.
- On-device image generation with local Stable Diffusion XL [21] or FLUX.1 [22] for offline operation and per-character consistency.
- Neural TTS (Tortoise, XTTS [24]) with cloned voices for character-specific narration.
- Music conditioned by current segment mood (per-segment crossfade) rather than per-genre static loops.
- Collaborative multi-user sessions with merged Q&A elicitation and turn-taking.
- Containerisation (Docker) + Redis sessions for distributed deployment.
- Automated evaluation: BLEU, chrF, MAUVE, narrative-arc detection for systematic benchmarking.
- Fine-tuned LoRA adapters per genre on curated open story corpora.

X. CONCLUSION

We have presented a multimodal interactive storytelling system built on a structured Q&A elicitation protocol, a modular backend, and provider-agnostic LLM inference. The system augments text with synthesised illustration, spoken narration and genre-conditioned ambient music, while remaining fully open-source and deployable on commodity hardware. Empirical evaluation across 50 stories and 15 participants demonstrates competitive narrative quality, sub-2-second cloud inference, and strong user satisfaction. We further introduced two end-user authoring affordances -- custom answers per question and custom user-added questions -- that we believe constitute a useful pattern for personalisation at the prompt boundary. The code-base is released at [27] to facilitate reproduction and extension.

XI. ACKNOWLEDGEMENT

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APPENDIX A. PROJECT STATISTICS

TABLE A1. Implementation footprint and evaluation scale.

Metric	Value
Python LOC	~3,800
JS / CSS / HTML LOC	~4,400
Backend modules	7 + main.py + config.py + run.py
DB tables	5
REST endpoints	14
Q&A questions	17 across 4 phases
LLM providers	4 (Ollama, Groq, HF, OpenAI)
Image generation	Pollinations.ai (free)
TTS	W3C Web Speech Synth
Ambient music	SoundHelix (8 genres)
Export formats	TXT, Markdown, HTML
Narrative phases	5 (Freytag)
Visualisations	2 D3 (Story Map, Char Graph)
Frontend features	14+ (themes, bookmarks, undo/redo, etc.)
Custom answer per Q	Yes
User-added questions	Yes (custom_q_N)
Stories evaluated	50
User study n	15

APPENDIX B. SAMPLE REST EXCHANGE

```

POST /api/story/generate
{
  "session_id": "a1b2c3d4-...",
  "responses": {
    "setting_1": "A magical fantasy realm",
    "char_1": "Elena",
    "char_2": "Brave and resourceful",
    "theme_1": "Fantasy",
    "plot_1": "Person vs. Supernatural",
    "custom_q_1": "Should the protagonist have an animal companion? -> Yes, a silver fox"
  }
}

200 OK
{
  "content": "In the realm of Aethermoor...",
  "word_count": 387,
  "segment_count": 1,
  "phase": "introduction",
  "choices": ["Explore further", ...]
}

```

APPENDIX C. REPRODUCIBILITY

All experiments were performed with the open-source release at [27], using Python 3.11.5 with fpdf 2.8.4, matplotlib 3.9.4 and numpy 2.0.2 on macOS 14. Image timing measurements were collected over a 100 Mbps residential link; readers replicating the study on slower links should adjust the serial-throttle constant correspondingly.